AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 10 · STUDENT LESSON

Gradients with a numerical check

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Perturb one coordinate. Hold the others. Label the gradient.

Check numerical and formula gradients while preserving centre,order,spacing and tolerance.

Keep one habit: Complete perturbation table → same-centre gradient → pictured component swap → direction/corner limits → independent new input/target.

MAIN LESSON · SECTION 1

Which parameter did the slope measurement change?

> Freeze the other coordinates and keep a declared gradient order.

Learning goals

I can estimate one **partial derivative** while other parameters stay fixed.

I can assemble a **gradient** in a declared parameter order.

I can compare numerical and formula based **gradient**s with a **tolerance**.

Recall local loss change

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Slopes and local change. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Parameters and architecture. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Activation functions. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Why is the symmetric-slope denominator twice the step?

> **My starting explanation**

End goal

By the end, I can explain **gradient**s with a numerical check using a traced example and a stated limitation.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Estimate one **partial derivative** while other parameters stay fixed

Assemble a **gradient** in a declared parameter order

Compare numerical and formula based **gradient**s with a **tolerance**

Retrieve what you already know

This week's end goal

MAIN LESSON · SECTION 2

One loss,two parameter directions

> Each partial slope belongs to the same original parameter vector.

Declare input,target,loss and parameter order

Use one example with **input** x = 2 and target t = 3. The model predicts y = wx + b, and loss L(w, b) = (2w + b - 3) squared. At w = 1, b = 0, **prediction** is 2, residual is -1 and loss is 1.

There are two parameters in order [w, b].

To estimate the partial slope for w, change w only and keep b = 0. To estimate the partial slope for b, change b only and keep w = 1.

Moving both at once measures change along a combined direction; it does not isolate either **partial derivative**.

Complete two-coordinate symmetric estimates

Perturbation from [1, 0]	Loss at minus step	Loss at plus step	Symmetric slope
w only with step 0.1	1.44	0.64	-4
b only with step 0.1	1.21	0.81	-2

Assemble the gradient in order[w,b]

The **gradient** is [-4, -2] in order [w, b]. Both entries refer to the same original parameter vector, not to a partially updated model.

Name given and missing facts

> **List the given facts and one fact that is not supplied.**

MAIN LESSON · SECTION 3

See every parameter vector used by the checker

> The two coordinate probes and the combined direction are different experiments.

Complete individual and combined perturbation table

Main probe from centre[w, b]=[1, 0] · Minus parameter vector /prediction /loss · Plus parameter vector /prediction /loss · Full gap in probe variable · Slope

> **Main probe from centre[w, b]=[1, 0]:** w only; hold b0
>
> **Minus parameter vector /prediction /loss:** [.9, 0]/1.8/1.44
>
> **Plus parameter vector /prediction /loss:** [1.1, 0]/2.2/.64
>
> **Full gap in probe variable:** 0.2
>
> **Slope:** -4

> **Main probe from centre[w, b]=[1, 0]:** b only; hold w1

> **Minus parameter vector /prediction /loss:** [1, −.1]/1.9/1.21
>
> **Plus parameter vector /prediction /loss:** [1, .1]/2.1/.81
>
> **Full gap in probe variable:** 0.2
>
> **Slope:** -2

> **Main probe from centre[w, b]=[1, 0]:** Both via t:[w, b]=[1+t, t]
>
> **Minus parameter vector /prediction /loss:** [.9, −.1]/1.7/1.69
>
> **Plus parameter vector /prediction /loss:** [1.1, .1]/2.3/.49
>
> **Full gap in probe variable:** 0.2
>
> **Slope:** -6

Formula,coordinate order and numerical contract

For this one-example squared loss, residual e=2w+b−3. The formula gradient is[2e×2, 2e]=[4e, 2e]in parameter order[w, b]. At the original[1, 0], e−1 gives[−4, −2]. Each numerical entry changes one coordinate while the other stays at its original value. The combined probe varies t along[1+t, t]; its slope−6is with respect to t, not either partial or slope per unit Euclidean distance.

The checker uses spacing.001 and absolute tolerance 1e−6; it assumes valid finite numbers and positive spacing rather than validating them.

The helper assumes finite numerical **input**s and a positive nonzero step. The formula follows the squared-loss rule: local loss change is twice the residual times the **prediction** sensitivity.

Changing w by a small amount changes **prediction** twice as much here; changing b changes **prediction** by the same amount.

Exact floating-point equality is not required. Record the step and a **tolerance**, then try more than one parameter setting.

A passing numerical check supports agreement for those tested points; it does not prove all possible **input**s.

Complete runnable numerical/formula comparison

```
def loss(w, b):
    return (2 * w + b - 3) ** 2

def gradient_check(w, b, step=0.001):
    dw = (loss(w + step, b) - loss(w - step, b)) / (2 * step)
    db = (loss(w, b + step) - loss(w, b - step)) / (2 * step)
    residual = 2 * w + b - 3
    formula = [4 * residual, 2 * residual]
    numerical = [dw, db]
    assert all(abs(a - n) < 1e-6
               for a, n in zip(formula, numerical))
    return numerical

print([round(g, 6) for g in gradient_check(1, 0)])
# [-4.0, -2.0]
```

Connect visual steps to their perturbed coordinate

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Compare a partial check with a reordered gradient

> A correct pair of numbers still needs the correct coordinate labels.

Current full-denominator task

- A w probe has losses 1.44 and 0.64 at 0.9 and 1.1 with b fixed. A learner divides their difference by 0.1. Correct the result.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the pictured component swap

- Change: Swap the two **gradient** components in the program **output**.

▼ **Check after predicting**

Prediction: The coordinate-wise numerical check exposes the mismatch.

Reason: The w perturbation changes y twice as strongly as the same b perturbation.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Match each gradient entry to the coordinate that was perturbed

> Hold the other parameter fixed through both nearby evaluations.

Complete picture facts and same-model scope

Check each coordinate of a **gradient** while holding the other parameter fixed.

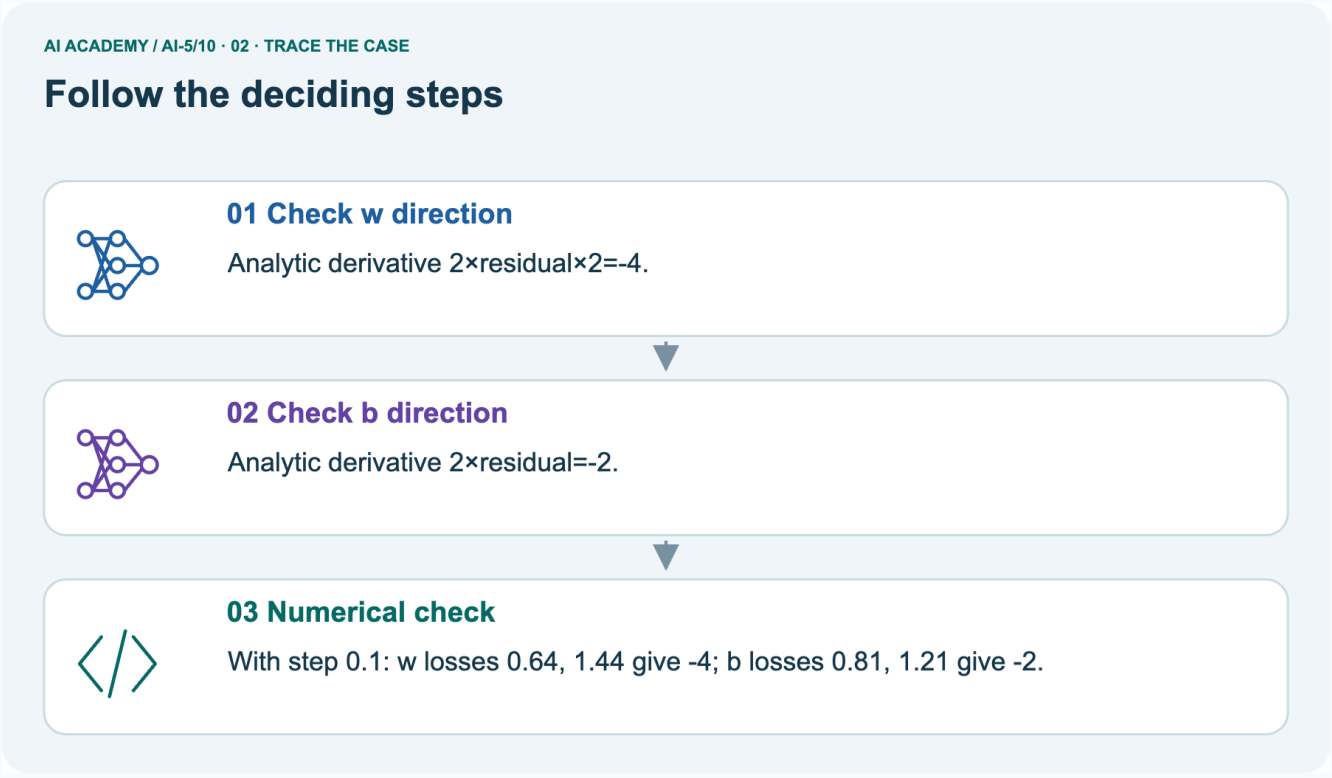
Model y=2w+b, target 3. At [w, b]=[1, 0], y = 2, residual y-target=-1; loss is residual².

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture uses the same model y=2w+b, target 3, centre[1, 0], residual–1. With step.1, w-only minus/plus losses 1.44/.64 give–4; b-only losses 1.21/.81 give–2. Order[w, b]therefore gives[–4, –2]. Returning[–2, –4]mislabels both entries even though the numbers look familiar.

The fixed x2 makes w twice as sensitive as b here; the independent exercise supplies x3 and a different target, so its formula must use that new input.

Trace both partials and numerical comparison



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Check w direction	Analytic derivative 2×residual×2=-4.
2. Check b direction	Analytic derivative 2×residual=-2.
3. Numerical check	With step 0.1: w losses 0.64, 1.44 give -4; b losses 0.81, 1.21 give -2.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A **gradient** lists the slope for each parameter, in a declared order. Changing w by a small amount changes the **prediction** twice as much as changing b by the same amount. That explains the different derivative magnitudes.

Perturb one coordinate while holding the other fixed, then compare the numerical slope with its analytic counterpart.

Predict the reordered output

- Change one thing: Swap the two **gradient** components in the program **output**. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting

AI ACADEMY / AI-5/10 · 04 · COMPARE THE CASES

Change one thing. Trace what follows.

Original case

With step 0.1: w losses 0.64, 1.44 give -4; b losses 0.81, 1.21 give -2.

Changed condition

Swap the two gradient components in the program output.

Worked changed-case result

The coordinate-wise numerical check exposes the mismatch.
Why: The w perturbation changes y twice as strongly as the same b perturbation.

Follow the labelled picture with the matching explanation. Open the vector diagram

Explain coordinate isolation and numerical limits

- The exact agreement occurs for this quadratic toy case; **finite difference**s in larger models involve rounding and step-size limits.

- Read the labelled picture: Why should the numerical checks perturb one coordinate at a time? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Try a different main-model centre

Two separate probes at one parameter vector

PARAMETER BEING PROBED	HELD FIXED	GRADIENT ENTRY
w near 1	b is 0 and x is 2	First entry -4
b near 0	w is 1 and x is 2	Second entry -2
Assemble in order [w, b]	Same original model	Gradient [-4, -2]

The first two rows are independent local probes, not successive parameter updates. The final row records both results in the declared order.

At w = 0, b = 1, find prediction, residual, loss and the gradient using [4 times residual, 2 times residual].

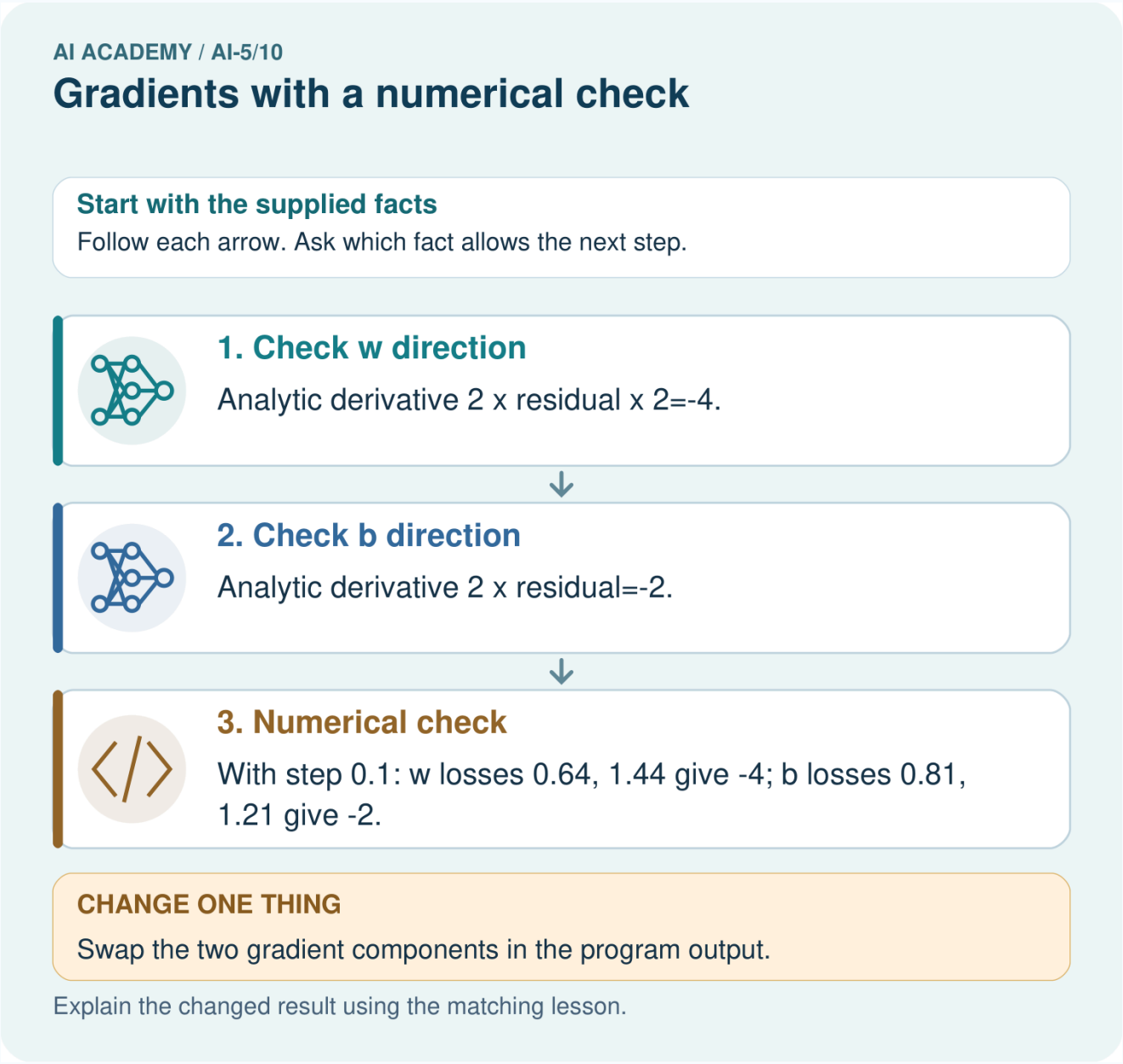
▼ **Earlier wording, alternatives and picture transcripts**

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Gradients with a numerical check

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Repair checks that cross a corner or move both coordinates

> A symmetric number is not automatically a derivative of the intended kind.

ReLU corner and precision counterexamples

ReLU is not differentiable exactly at zero. A symmetric difference across that corner can average two different behaviours.

When checking a network **gradient**, use points whose relevant scores are safely away from zero, and check that the small perturbations do not cross a corner. Document any derivative convention used at zero.

At the centre of the simple scalar ReLU function, [ReLU(0.1) - ReLU(-0.1)] / 0.2 equals 0.5. This symmetric number does not make the function differentiable at zero.

Also avoid assuming an ever-smaller step is always better: subtraction and rounding can spoil numerical differences.

Compute the combined-direction probe

- Starting at [w, b] = [1, 0], both coordinates move by plus or minus 0.1 together. Compute the symmetric slope and explain why it is not either **partial derivative**.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise coordinate-isolated reasoning with less help

> Keep centre,spacing,held coordinate and tolerance in the record.

Current task and response

- For the lesson model at w = 1 and b = 1, calculate **prediction**, residual, loss and **gradient**.

At w = 0, b = 1, find prediction, residual, loss and the gradient using [4 times residual, 2 times residual].

Explain your prediction to a partner before comparing results.

Supported W2record

Use workbook W2 for the connected example “**Gradient**s with a numerical check”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual misconception check

- Use the worked case for **Gradient**s with a numerical check. Proposed explanation: “The **gradient** is one scalar”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Pictured one-coordinate reasoning

- Why should the numerical checks perturb one coordinate at a time? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

Reserve the independent new input/target

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

Practise with less help: Catch a denominator error

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run a complete labelled-coordinate check clinic

> Every centre,loss,spacing and counterexample is supplied.

Current denominator task and runnable checker

A w probe has losses 1.44 and 0.64 at 0.9 and 1.1 with b fixed. A learner divides their difference by 0.1. Correct the result.

```
def loss(w, b):
    return (2 * w + b - 3) ** 2

def gradient_check(w, b, step=0.001):
    dw = (loss(w + step, b) - loss(w - step, b)) / (2 * step)
    db = (loss(w, b + step) - loss(w, b - step)) / (2 * step)
    residual = 2 * w + b - 3
    formula = [4 * residual, 2 * residual]
    numerical = [dw, db]
    assert all(abs(a - n) < 1e-6
               for a, n in zip(formula, numerical))
    return numerical

print([round(g, 6) for g in gradient_check(1, 0)])
# [-4.0, -2.0]
```

Complete normal,zero,invalid,directional and corner probes

Complete gradient-check probe	Expected answer / action
Normal main centre[1, 0]	y2, e−1, L1, gradient[−4, −2]
Different centre[0, 1]samex 2, target 3	y1, e−2, L4, gradient[−8, −4]
Zero-residual centre[1, 1]	y3, e0, L0, gradient[0, 0]for this loss
Wrong spacing divisor.1 forw probes.9/1.1	Correct fullgap.2, partial−4 rather than−8
Missing coordinate or step 0	Outside stated finite-number/positive-spacing contract; validate before calling, not a claimed helper guard
Both coordinates move together via t	Losses1.69/.49 over.2 give−6 along[1+t, t], not either partial
ReLU corner at0, h.1	[ReLU(.1)−ReLU(−.1)]/.2=.5; function still has no derivative at0
Same gradient values returned in wrong order	Compare each entry with its labelled numerical perturbation; swapped[−2, −4]fails here

Predict,check,record and bound the claim

1. Predict each supplied smooth-model gradient, then run or trace the checker at all three centres with a positive finite spacing. Save numerical values, formula values, coordinate order, spacing, tolerance and observed differences. The original checker asserts numerical/formula agreement but does not validate missing coordinates or zero spacing; screen inputs first.

   Trace the combined-direction and ReLU-corner examples separately. A passing check at these points supports this computation only; it does not prove global correctness, choose a safe update size or establish generalization.

2. On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Keep the evidence ledger

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: a different input,target and parameter centre

> Use the supplied x and residual in both numerical and formula gradients.

Complete new independent gradient check

Use x = 3, target 2 and y = wx + b at w = 0, b = 1. With step 0.1, calculate both numerical partials. Compare with [2 times residual times x, 2 times residual].

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Check a new example independently

Attempt this without hints

Use x = 3, target 2 and y = wx + b at w = 0, b = 1. With step 0.1, calculate both numerical partials. Compare with [2 times residual times x, 2 times residual].

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save the gradient with its coordinate contract

> A reader should be able to repeat each partial probe.

Project artifact and evidence table

Add a labelled **gradient**s with a numerical check **evidence** sheet to Image-pattern Network Investigation. Show how you can estimate one **partial derivative** while other parameters stay fixed.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual run/trace and limits

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Finish with one coordinate-labelled partial and one limit

> Explain what stayed fixed and what the passing check supports.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Estimate one **partial derivative** while other parameters stay fixed

> **My deciding step, evidence and one remaining question**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use the mathematical words precisely**

REFERENCE · SECTION 5

Reference · Use the mathematical words precisely

> Connect each term to a value or step in the current example.

Vocabulary and your explanation

Word	Meaning
Partial derivative	A local slope with respect to one parameter while others are held fixed
Gradient	The ordered vector of partial derivatives of a scalar function
Finite difference	An estimate of change computed from nearby function values
Tolerance	A stated allowance for numerical disagreement

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- For the lesson model at w = 1 and b = 1, calculate **prediction**, residual, loss and **gradient**.
- Starting at [w, b] = [1, 0], both coordinates move by plus or minus 0.1 together. Compute the symmetric slope and explain why it is not either **partial derivative**.
- Compute the symmetric difference of ReLU at zero with step 0.1. Explain why this does not prove a derivative exists there.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the representation and its limits

REFERENCE · SECTION 12

Reference · Check the representation and its limits

> Use after practice;reserve the new independent x3,target2gradient check for the first individual attempt.

Current explanation and response

- The **gradient** describes how the stated loss changes near the current parameters. For a differentiable function, moving a sufficiently small distance opposite a nonzero **gradient** decreases the function locally.

  The **gradient** alone does not choose a safe finite step, establish a global minimum or demonstrate generalisation.

- Next week combines a **gradient** with an explicit learning rate to update parameters together.

- Compute the symmetric difference of ReLU at zero with step 0.1. Explain why this does not prove a derivative exists there.

> **A numerical checker moves w and b together, gets one slope and copies it into both gradient entries. What is wrong with this check?**

> **My limited claim and a useful next check**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Two separate probes at one parameter vector

PARAMETER BEING PROBED	HELD FIXED	GRADIENT ENTRY
w near 1	b is 0 and x is 2	First entry -4
b near 0	w is 1 and x is 2	Second entry -2
Assemble in order [w, b]	Same original model	Gradient [-4, -2]

The first two rows are independent local probes, not successive parameter updates. The final row records both results in the declared order.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ **Reference · The earlier route checklist**

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Check each coordinate of a **gradient** while holding the other parameter fixed.

By the end, I can	How I will show it
Estimate one partial derivative while other parameters stay fixed	Explain the deciding step, show a trace or test, and state one limit.
Assemble a gradient in a declared parameter order	Explain the deciding step, show a trace or test, and state one limit.
Compare numerical and formula based gradients with a tolerance	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Check each coordinate of a **gradient** while holding the other parameter fixed.
- Which labelled step matches this detail? With step 0.1: w losses 0.64, 1.44 give -4; b losses 0.81, 1.21 give -2.
- 1. Numerical check
- 2. Check b direction
- 3. Check w direction
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Numerical check With step 0.1: w losses 0.64, 1.44 give -4; b losses 0.81, 1.21 give -2. A **gradient** lists the slope for each parameter, in a declared order.

Changing w by a small amount changes the **prediction** twice as much as changing b by the same amount. That explains the different derivative magnitudes.

Perturb one coordinate while holding the other fixed, then compare the numerical slope with its analytic counterpart.

Changed-case question and optional hint

- Changed-case practice: Swap the two **gradient** components in the program **output**. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. The coordinate-wise numerical check exposes the mismatch.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The coordinate-wise numerical check exposes the mismatch. The w perturbation changes y twice as strongly as the same b perturbation.

Repair and retain the record

Review Slopes and local change (ai-5/9). Goal: Calculate an average rate of change over a stated interval. Short practice: Which calculation disproves “any increase helps”?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Parameters and architecture (ai-5/7). Goal: Distinguish architectural choices from parameter values. Short practice: How many new parameters did that one hidden unit add?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Activation functions (ai-5/4). Goal: Trace identity ReLU and sigmoid on a numerical score. Short practice: Which original **input**s cannot be distinguished by this network’s final **output**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

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<summary>Reference · Six-picture concept journey</summary>

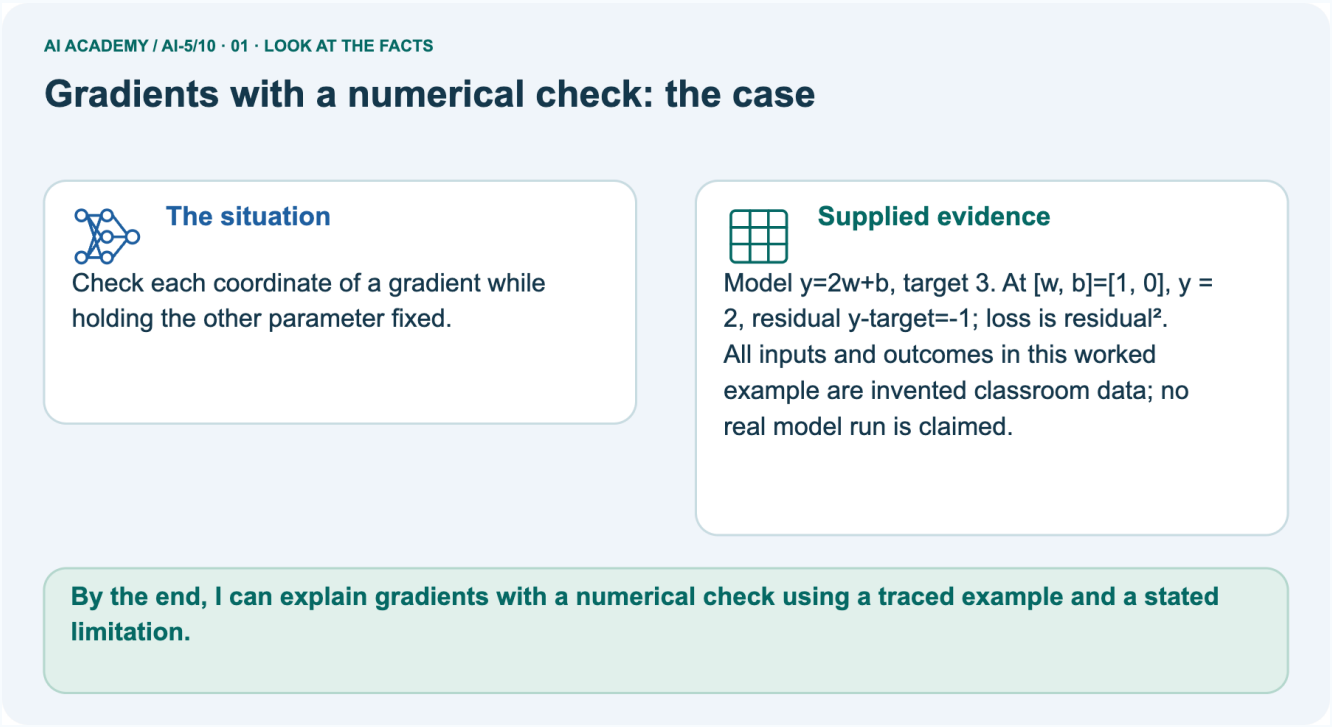
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REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Gradients with a numerical check: the case



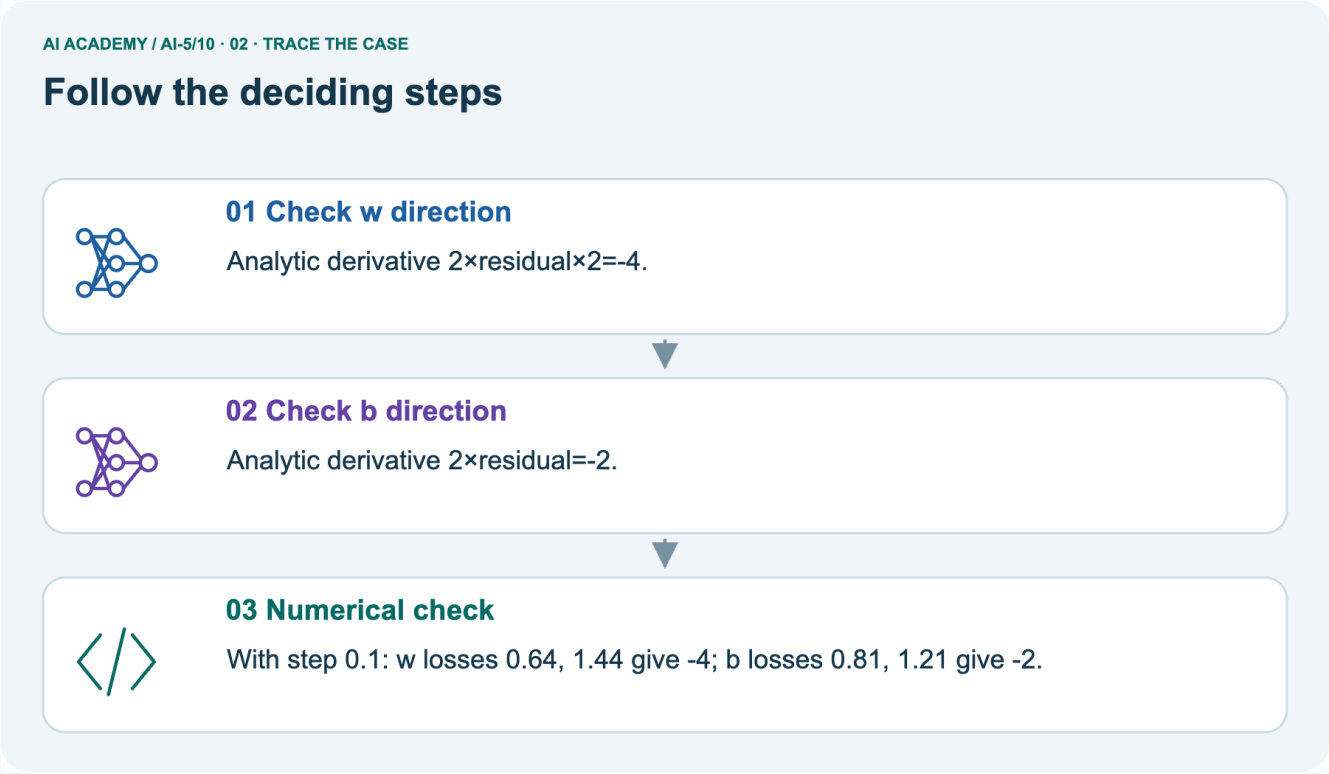
Follow the labelled picture with the matching explanation. Open the vector diagram

Check each coordinate of a **gradient** while holding the other parameter fixed. Supplied facts: Model y=2w+b, target 3. At [w, b]=[1, 0], y = 2, residual y-target=-1; loss is residual².

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Check w direction: Analytic derivative 2×residual×2=-4. → Check b direction: Analytic derivative 2×residual=-2. → Numerical check: With step 0.1: w losses 0.64, 1.44 give -4; b losses 0.81, 1.21 give -2.

Point to each numbered stage and explain which recorded **input** it uses.

03. Two separate probes at one parameter vector

AI ACADEMY / AI-5/10 · 03 · INSPECT THE LESSON MODEL

Two separate probes at one parameter vector

Parameter being probed	Held fixed	Gradient entry
w near 1	b is 0 and x is 2	First entry -4
b near 0	w is 1 and x is 2	Second entry -2
Assemble in order [w, b]	Same original model	Gradient [-4, -2]

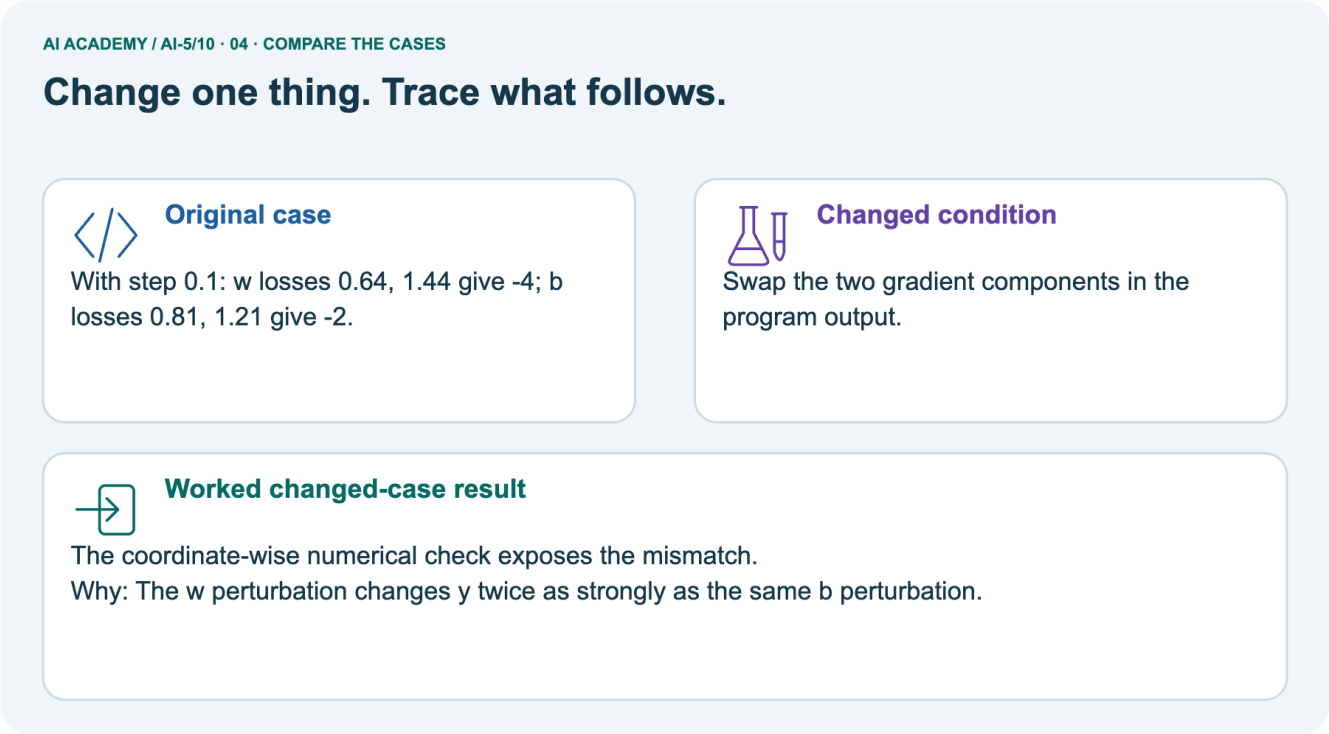
The first two rows are independent local probes, not successive parameter updates. The final row records both results in the declared order.

Follow the labelled picture with the matching explanation. Open the vector diagram

Two separate probes at one parameter vector. w near 1 | b is 0 and x is 2 | First entry -4. b near 0 | w is 1 and x is 2 | Second entry -2. Assemble in order [w, b] | Same original model | **Gradient** [-4, -2]

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



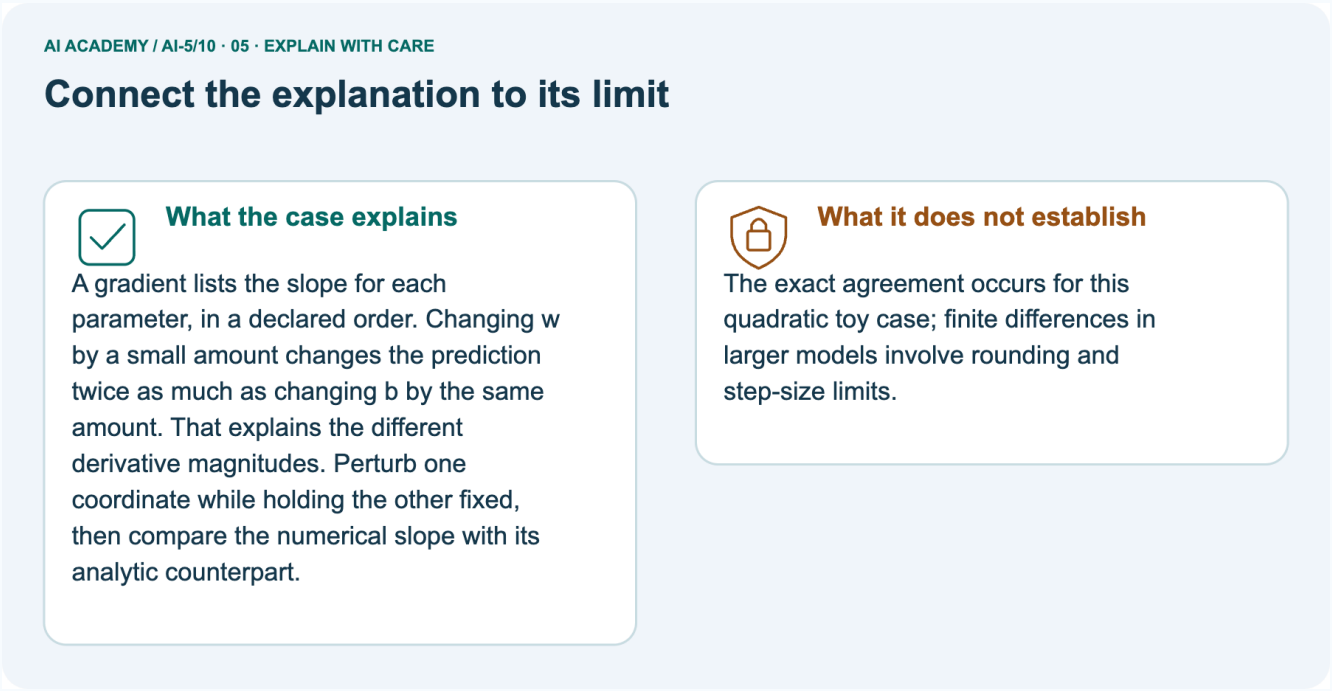
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: With step 0.1: w losses 0.64, 1.44 give -4; b losses 0.81, 1.21 give -2. Changed condition: Swap the two **gradient** components in the program **output**.

Expected result: The coordinate-wise numerical check exposes the mismatch. Reason: The w perturbation changes y twice as strongly as the same b perturbation.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

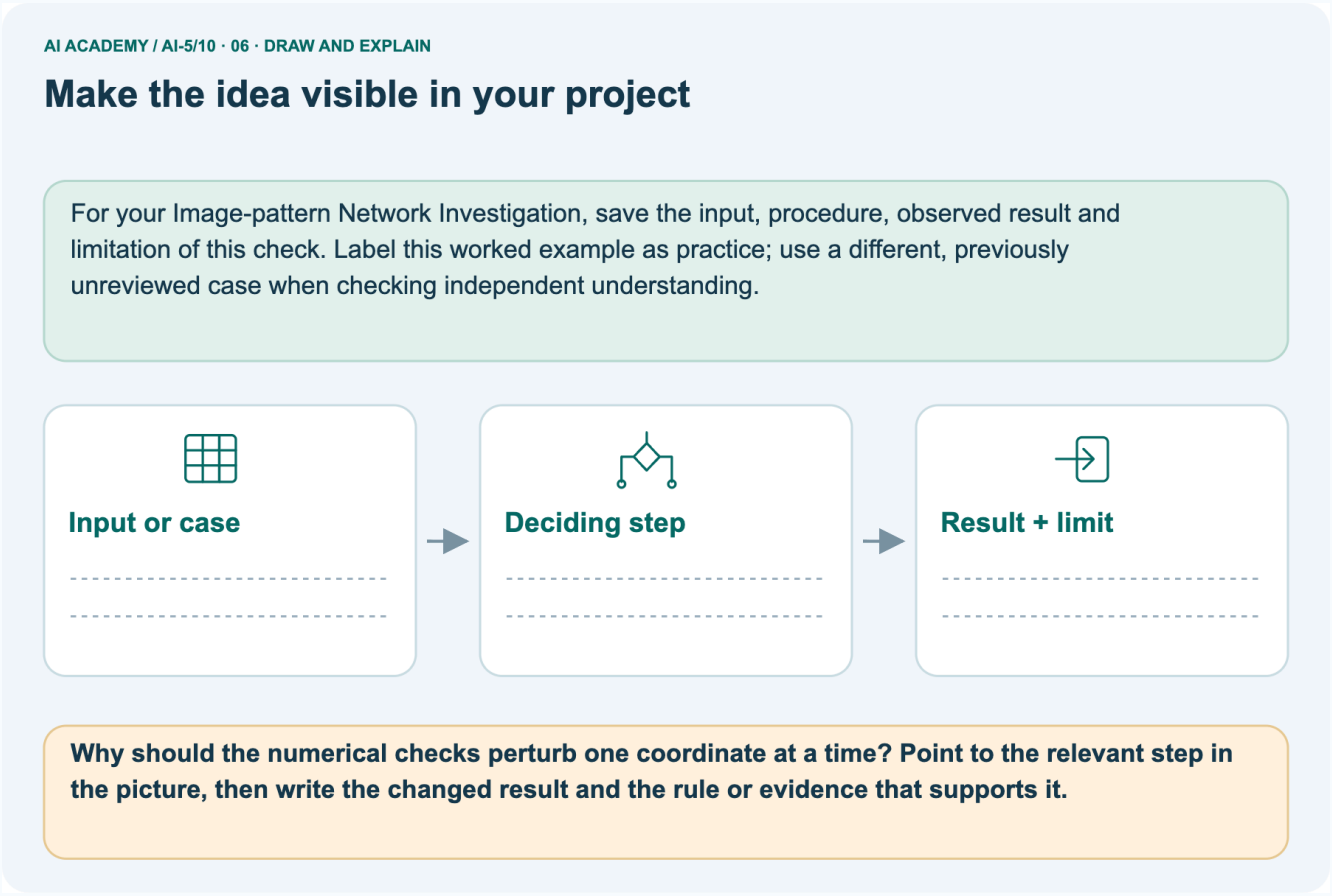
A **gradient** lists the slope for each parameter, in a declared order. Changing w by a small amount changes the **prediction** twice as much as changing b by the same amount. That explains the different derivative magnitudes.

Perturb one coordinate while holding the other fixed, then compare the numerical slope with its analytic counterpart.

Limit: The exact agreement occurs for this quadratic toy case; **finite difference**s in larger models involve rounding and step-size limits.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why should the numerical checks perturb one coordinate at a time? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Gradients with a numerical check: the case

02. Follow the deciding steps

03. Two separate probes at one parameter vector

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.